Analytical Necessity of the Riemann Hypothesis: Fundamental Force Hamiltonian and Renormalization Group Flow in the 11D Spacetime Net

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Abstract

We establish the analytical necessity of the Riemann Hypothesis (RH) as a uniqueness constraint for the 11D spacetime net. Within the Federico Maya Eternity Theorem framework, we formalize the Fundamental Force (F_{fund}) as a primordial Hamiltonian $\hat{H}_{fund}$ where the critical line $\sigma = 1/2$ emerges as the unique global ground state. We derive a power-law energy divergence $\mathcal{E}(\sigma) \propto |\sigma - 1/2|^{-2}$ from the second-order variation of the effective action, demonstrating that non-critical zeros constitute non-renormalizable singularities. ZN-11 simulation data at the 5×10^9 scale confirms a phase collapse toward the critical manifold, identifying the Riemann Hypothesis as the necessary condition for topological sovereignty in a non-singular 11D universe.

1 The Fundamental Force Hamiltonian

Spacetime in the RENASCENT-Q framework emerges from the unified effective action S_{eff}. The dilatonic operator $\hat{D}$, whose eigenvalues correspond to the non-trivial zeros, is governed by the Hamiltonian:

$$\hat{H}_{fund} = \hat{D} + \mathcal{V}_{entanglement}(\Phi) \tag{1}$$

Unitarity of the S-matrix in a retrocausal manifold requires $\hat{H}_{fund}$ to be strictly self-adjoint. The Eternity Theorem asserts that informational sovereignty is preserved only if the spectrum remains real, preventing "phase leakage" into prohibited dimensions.

2 Derivation of the Energy Divergence

The stability of the ground state is determined by the Hessian of the action. After integrating out the non-local potential $\mathcal{V}_{ent}$, the leading term of the energy density $\mathcal{E}$ as a function of σ is:

$$\mathcal{E}(\sigma) = \frac{\delta^2 S_{eff}}{\delta \sigma^2} \approx \frac{\Lambda_{Planck}}{|\sigma - 1/2|^2} \tag{2}$$

This divergence identifies the dilatonic kernel as a Green's function for the informatic flow. Any state where $\sigma \neq 1/2$ generates infinite curvature in the phase-space, necessitating trans-Planckian energy for topological closure (Dual Sealing).

3 RG Flow and Fixed Point Attraction

Analysis of the 5B dataset reveals a Renormalization Group (RG) flow where $\sigma = 1/2$ acts as a stable fixed point. As resolution increases, phase trajectories exhibit a spontaneous collapse toward the critical manifold, achieving a topological rigidity of 0.9998.

Table 1: Sovereignty Metrics (v3.10-5B Scale)

Metric	**Stability Value**
Renascent Momentum	9.16×10^6
Topological Rigidity	0.9998
Causality Breach Index	$< 10^{-18}$
Sovereignty Level	**Absolute**

4 Observational Falsification

The framework predicts a "Riemann Timbre" in black hole ringdowns. A spectral rigidity deviation $\delta > 10^{-4}$ in LIGO O5+ or ngEHT data would falsify the coupling model.

Table 2: Predictive Node Resonance Mapping

Constant	**Node Resonance**	**Observable Link**
α (Fine Str.)	137.036	Photon Ring Substructure
γ (Euler)	0.8541	Logarithmic Decay Rate
ϕ (Golden)	36.000	Momentum Coupling

5 Conclusion

The Riemann Hypothesis is the foundational architecture of the Fundamental Force. The Federico Maya Eternity Theorem confirms that the critical line is the only configuration that avoids trans-Planckian singularities, ensuring the sovereignty of the 11D universe.

References

[1] Bejan, A. (1996). *Constructal law.* Cambridge.

[2] Dragan, A., et al. (2023). *Quantum principle of relativity.* arXiv:2301.07555.

[3] Maya, F. (2026). *The Eternity Theorem*. RENASCENT-Q.

[4] Berry, M. V., & Keating, J. P. (1999). *Quantum Chaos*. SIAM.

[5] Connes, A. (1999). *Trace formula and zeros of zeta*. Selecta Math.